

Decision Tree Regression Analysis

Variance Reduction Criterion (2 Iterations)

1 Dataset and Baseline Metrics

The following dataset contains real estate data where **Squarefeet** is the predictor (X) and **Price in Lakhs** is the target variable (y).

ID	Squarefeet (X)	Price (y)
1	800	50
2	850	52
3	900	54
4	1500	90
5	1600	95
6	1700	100

Parent Node (Baseline):

- **Total Samples (N):** 6
- **Parent Mean (μ):** 73.5
- **Parent Variance (S^2):** 471.92

2 Iteration 1: Root Split Candidates

We test candidate split thresholds by calculating the Weighted Variance and the resulting Variance Reduction for each possible split.

Threshold	L. IDs	L. Mean	L. Var	R. IDs	R. Mean	R. Var	W. Var	Var. Red.
SqFt $\leq$ 825	[1]	50.0	0.00	[2,3,4,5,6]	78.20	433.76	361.47	110.45
SqFt $\leq$ 875	[1,2]	51.0	1.00	[3,4,5,6]	84.75	327.69	218.79	253.13
SqFt $\leq$ 1200	[1,2,3]	52.0	2.67	[4,5,6]	95.00	16.67	9.67	462.25
SqFt $\leq$ 1550	[1,2,3,4]	61.5	272.75	[5,6]	97.50	6.25	183.92	288.00
SqFt $\leq$ 1650	[1,2,3,4,5]	68.2	397.76	[6]	100.00	0.00	331.47	140.45

Table 1: The split at **Squarefeet** $\leq$ 1200 yields the highest Variance Reduction (462.25) and is selected as the root split.

3 Iteration 2: Splitting the Child Nodes

3.1 Left Child Node (IDs 1, 2, 3 | Parent Variance = 2.67)

Threshold	L. IDs	L. Var	R. IDs	R. Var	W. Var	Var. Red.
SqFt $\leq$ 825	[1]	0.00	[2, 3]	1.00	0.67	2.00
SqFt $\leq$ 875	[1, 2]	1.00	[3]	0.00	0.67	2.00

Table 2: Both splits yield identical reduction; **Squarefeet $\leq$ 875** is selected.

3.2 Right Child Node (IDs 4, 5, 6 | Parent Variance = 16.67)

Threshold	L. IDs	L. Var	R. IDs	R. Var	W. Var	Var. Red.
SqFt $\leq$ 1550	[4]	0.00	[5, 6]	6.25	4.17	12.50
SqFt $\leq$ 1650	[4, 5]	6.25	[6]	0.00	4.17	12.50

Table 3: Both splits yield identical reduction; **Squarefeet $\leq$ 1550** is selected.

4 Final Leaf Nodes Summary

After 2 iterations, the decision tree is fully constructed with the following 4 leaf nodes:

Leaf Condition	Sample IDs	Prices	Mean Predicted	Final Var
Squarefeet $\leq$ 875	[1, 2]	50, 52	51.0	1.00
875 < Squarefeet $\leq$ 1200	[3]	54	54.0	0.00
1200 < Squarefeet $\leq$ 1550	[4]	90	90.0	0.00
Squarefeet > 1550	[5, 6]	95, 100	97.5	6.25

5 Decision Tree Diagram

